ActInf Textbook Group ~ Cohort 7 ~ Session 6 (Chapter 3) ~ 8/26/2024

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okay welcome back Arts of cohort 7 um
we're in
Three
so beginning in
three is there anything that anybody
wants
to begin with like any
overall thought on three or any any
place to start
I guess I was a bit a bit slow with the
uh with the recap of the the chapter
here but one one thing at least that I
uh thought a bit about while reading it
was and I guess uh not unique to this
chapter but the proceeding ones as well
is the interesting connection
to other
domains and mention like the the physics
and activism and cybernetics and all of
these different ideas and also like how
it seems as if active inference has
evolved over time and emerged from all
of these ideas and like the connection
between it which I U I I think seems
very I think it's very interesting and
um I was just wondering if there are any
Resources with with someone that has
like uh mapped up
the connections and perhaps like
Evolution
over of of the ideas over over in
history uh so that you can get like a
good overview of of of how how this has
evolved and it's connected to other
areas and perhaps like even what

different assumptions and applications
go into
it yeah that's a great suggestion I mean
the
you mentioned kind of like a um
historiography like bibliography tracing
but then also like the claims and the
rhetorical assertions and understanding
what which
claims um
transferred it also kind of reminded me
of
this um I I'll I'll put it here but this
in the complexity
Sciences this
one graph I mean even even though it's
just one partial representation becomes
quite
popular even though it's just a
subsample of the tip of the iceberg of
one person's representation of one way
to view it but but then it does give
some
coherence um even just from chapters 1
and 10 of the
textbook it could be possible to um pull
out some of these kinds of threads and
then yeah that Beal and it's like how
different would it really be from active
inference mathematics of
complexity systems and complex systems
Network
science cybernetics maybe there's
information Theory here
somewhere artificial
intelligence so agent based
modeling so I think it would it would it
would have a lot of similarities to this

but but I I agree
yeah yeah super cool I'll check that
link out um I guess it has some some
resources resources in the in the book
like the table 3.1 for instance but also
from like chapter 2 the
figure
um which one is it 2.6 as well which I
think have like quite nice connections
to
one um kind of
uh history crossover moment from
uh from the last
Friday
this uh model stream 13 synthesizing the
born rule with reinforcement
learning so they're coming from a um
Quantum measurement empiric IAL setting
and then as they started to relax
certain assumptions and try to
generalize the born rule they got into
the generalized sensory motor
interface and the sort of generalized
cognitive
setting so that's like a low road
generalization and so we had some
interesting
um crossover moments were like they were
they're in the labor
measuring these Quantum
collapses and then they've converged
upon where from active
inference it it generalized into the
quantum over the last several years but
then so there could be cool ways to
go
[Music]
um any other random thoughts

on three or or the high
road I mean that's kind
of what three is about one just opens
the book just like 10 closes the book
two took us on the low
road now three is going to start from up
here and come back down to active
inference so that we're we we get off
the slide of chapter three and we end up
right a chapter four where we finally
see the active inference generative
model but yeah any thoughts on high road
three like just to kind of illustrate
that um yeah yeah good good the um to me
I mean it's almost like we're just
creating a
more granular structure around the
Socratic method
how
so uh we've got you know agents in an
environment and they're um asking
questions about the environment and
receiving a response from their
environment and then um integrating that
response into their internal model of
the world and rinse and repeat
yes
interesting there's the there's the
socratic element on that there's the
hegelian element on that and like all of
these more like composite
Frameworks where like it doesn't seek to
reduce situation down
to just like a a a monad Unity but to a
DI dialectical
Unity are there other aspects to the
Socratic
method Wonder reflect refine and

cross-examine restate
repeat this this connects I mean to the
active inference
ontology what is surprise
or you know these are these are the
kinds of
questions and these are like a
Playbook what does the book mean
by
novelty what why would they think that
the K Divergence is the way to do this
is the K Divergence the only way to do
this
there's a nice buildup towards
three in chapter one this is kind of the
minimal agents Niche figure ground like
two-part um
partition that gets nuanced into kind of
like a four
state partition we have a a true
internal and external and then there's
two aspects to the blanket or you can
kind of collapse those two into to one
blanket and that's where the the Markov
blanket concept initially was was just
like it was one blanketed node and then
one of friston's Novelties to to adapt
the Markov blanket concept into the
cybernetic setting was to kind of split
that blanket or interface into an
inbound and an outbound so that's where
we get the
four but here it's still framed as like
they're just four variables like in a
program in chapter 3 it picks up with
those
four and now they're defined in terms of
flows like the dot being a

derivative so now there's like four
simultaneous flows
on this fourfold partition so that that
is coming more from the basian mechanics
but that just shows like it's like
chapter one is the is the parent of both
two and three and then chapter two goes
more of like an engineering program way
of doing this like well what are the
variables like what are like um what's
the local computer State the remote
computer State and then the the the
sending to and from State here's more of
that physics um way to see
it we we can look at this figure in
context but this is kind of like showing
uh B is the blanket X is for external mu
is for internal kind of like I maybe mu
for mind for M or something but it's
like saying if you have like a um just
just to make a b numbers if you have a 0
five correlation between
um internal and blanket and a 05
correlation between blanket and external
then you're going to end up with a 0.25
correlation between internal and
external just because it's kind of like
it's two hops and each one has a 0 five
correlation so that's kind of how you
end up with correlations across blanket
because of kind of like the chain rule
of partial correlations
any other random thoughts on three or we
can look at the accident
questions or we can look at the text
this is one of oh yeah go ahead Jeff
um the one thing that I
think and again you know we being stuck

here in the in the forward era of time
um Melvin vosen um has developed the
theory of info Dynamics which is a a
corollary to thermodynamics and I feel
that there's overlap
here and uh I can't I can't be more
specific than that but I can link the
the the
article
um and I don't know maybe somebody wants
to take this and run with it or maybe I
will at some
point okay I remember Alli
in a um characteristically um oi
styled Terrence Deacon has also has
touched upon the info
Dynamics it's kind of like the the data
information knowledge
wisdom like kind of but but it's it's a
little bit different than that but it's
like what are sort of the um
syntactic dynamic
of of information like at the sensor
processing level then as you move into
more semantic levels of
informations um how does and then and
then Deacon highlights that um those
higher
levels they they are dealing more and
more explicitly with like adjacent
posses rather than processing what is
like immediately there like you can't
understand the higher levels without
also considering like this kind of like
as yet unmanifest shell of which is kind
of like when we get into action planning
That's a classic example like most plans
will not happen yet the expected free

energy still kind of considers
them but in so in this question we'll
see if the qu if if the actual maybe the
real questions will be
address but this image this is from one
of the
earliest like
pre-2010 winged
snowflake so here temperature increasing
is going down so it's colder higher up
in this image and this is like the
actual sky so it's like it's colder
higher up and then like closer to the
ground it's cooler so there's some
critical
temperature under which from an
elevation perspective like the snowflake
is going to
melt Above This level it will persist as
a
snowflake so if it was a inactive
Snowflake and um then let's say you
started with a cloud of like a bunch of
snowflakes that was that were high up um
then you just have would have diffusion
so you wouldn't need to propose any kind
of action selection it would just be
like um if there was lamin or flow all
the snowflakes would stay up here if it
was going straight down all of them
would fall down like that you wouldn't
falsify the hypothesis that it was just
like a particle but then what you see
with like um life is you see
action that that acts as if I mean if
you came back millions of years
later then you you'd be seeing
snowflakes that would be acting as if

they were staying above that temperature
threshold because the ones that didn't
have kind of crashed
out so this is like showing up in a
population of actual neant Tropic
organisms and they've been
fighting um death you know even without
gting reproduction stuff like that but
but that that was an early physics
inspired way to think about
survival but this this could be a blood
sugar critical threshold temperature
threshold okay quick yeah that's that's
a fun one of the early fun
representations um where was that from
um you
know let's just look up what it's like
it maybe 2006 first in paper okay okay
cool
not that kind of
[Laughter]
snowflake 2000 I can look look it up
yeah yeah
yeah yeah this this 2006 paper with
Kilner and
Harrison yeah here's
the here's with a different notation
but but we see a lot of the
same input and output and then here's
like three levels like this is kind of
the fast inference
layer then there's the slower
attentional neurom modulation layer and
then there's an even slower topological
change neuronal connections so it's like
already there's kind of nested time
scales of
inference um because those were already

being considered in the
fmri SPM setting let's see what else is
in this
paper this kind of looks like a heat
engine kind of like a a duty cycle on an
engine cortical
columns comes back in the
book generative and recognition
directions from pixels to planning
that's like the most recent um paper but
I mean here we have
pixels and still that same kind of um
minimal
case already kind of
highlighting uh FM and repetition
suppression and those are some of the
robust EEG and the fmri type findings
like in different populations of people
and then like when the mismatch
negativity and the 200 millisecond tone
and all those kinds of
things yeah pretty interesting almost 20
years
okay F first question is about a Markoff
blanket in
time okay a as usual attaching a last
name to something does not always add
information because it doesn't do
anything other than just make sort of a
historical illusion but this is the
marvian property in time which is
basically that the past only influences
the future through the present so it's
like the current state of the program
influences the next state and anything
prior to that gets distilled or like
projected onto the present State um
that's the that's a marov chain and then

if there are like two time states of
time
dependency then it could be like a one
marov chain with two time steps like if
your last two moves were up up then
here's where that transitions to so even
if it is like multiple time steps in the
simulation you can still kind of model
that as a one Markov chain or you can
you can model it with with multiple time
steps but then people also have have
broader questions like you know without
knowledge of the present what what can
you do say
two what is a statistical rather than a
non-statistical
interaction
um we're talking about a statistical
model so all the interactions which look
like edges on a base graph are
statistical
interactions um then it's an interesting
question what what would make a
non-statist maybe that's the real quote
real interaction
itself and then third
question why does Independence between
the flows on the two side of the blanket
matters so much
it's a complex question like matters so
much relative to
what internal and external States they
can influ you can set it up so that they
have total influence on each other so it
isn't that it's that the blanket
isolates it it defines how they
articulate and then it could depend a
lot or not but I think a key piece is

that their random fluctuations are kind
of buffered from each
other like they're getting the noise is
getting
um at least the noise or orthagonal
they're being like randomly sampled from
noise but that may or may not matter in
a certain situation
these are great
um
questions there's kind of this these
cousins of of closely related
terms like of course we have active
inference but then we also hear about
Basi and brain hypothesis that's in
1.2 and then there's predictive
processing predictive coding um this was
question that we asked of friston in
and I think the answer has also kind of
sharpened since
then um you could have a situation where
all of these were in play you could have
an active inference
system that was brain likee being
described by basian brain
hypothesis where the processing that it
was doing was based in terms of
prediction and anticipation and and all
of that and
the messages being passed between layers
reflected predictions and their
discrepancies so all four of these could
be in
play or you could have situations that
kind of nuanced that um but they're more
similar than
not predictive processing highlights
making predictions which is

is almost without a space between with
the basian brain hypothesis because
that's kind of what a prior observation
R predictive
coding sometimes is used just
equivalently to predictive
processing other times and this is
explored a little bit with Maria in um
the live stream
43.0 and she studied some of the
philosophy of the predictive processing
predictive processing can refer to like
the fact that the whole cognitive stack
does this predictive process
whereas predictive coding is more of a
subhypothesis that the messages being
that are transpiring between layers are
encoded in terms of the
predictions but some other type of
message format could be
passed active inference amongst all of
these highlights like the role of action
whereas basian brain hypothesis it could
be a passive basian brain but again it
could be an active one and predictive
processing predictive coding a lot of
those models were sensemaking only but
you could consider making predictions to
be like a kind of internal action but
then those types of models generally
earlier were not considering like I
movements but once you get into the
sense making for what the retina sees in
a moment and the choice of not just what
to predict at the next time step but
where to move your eyes and what to
predict when you move your eyes there
then that gets more into the full active

and front
setting so yeah that that's a question I
thought about many times so it's great
to get some more meat on that but I
specifically wondered about the uh like
overview picture or over overview figure
of the low road on the high road where
predictive coding is specifically on the
the low road and predicted processing on
the the high road if I didn't mix them
up now
um could you perhaps explain
why yeah why they're separated like that
uh let to put some more meat on it sure
no I mean it's it's a it's a great
observation predictive coding down here
yeah first off um unlike a subway map
it's it's not that these are like
strictly prerequisite in a certain order
yeah so but so it's kind of interesting
but you know it's so it's not that this
is like the only final representation
but um predictive coding it could be
argued it's an
information optimal way to send messages
about
predictions within a
system
so when doing this kind of perceptual
inference it's like let's just say that
we were we were um we we were um doing a
measurement scale and and uh we're
measuring cars they're 1,000
lb we expect them to be 1,000
lb we could um perhaps most
parsimoniously report like minus three
if it was three below or plus seven so
it's like that's smaller than saying

17 so th this could be as argued as kind
of just like a
model as a sort of like an intra model
strategy for information theoretic
optimal transmission of information
predictions whereas predictive
processing one which it's I don't think
it'd be um illit on the low road because
we're talking about basing brain and
stuff but one reason why it could be put
on the high road High Road aims to start
with minimal assumptions other than a
persistence
either a self-observed persistence like
survival or a sort of measurement type
persistence like a experiment or
observing something
repeatedly so in that situation you
don't know what is happening
necessarily amongst the elements of what
you're observing but you could say there
it it evinces behavior that it is doing
something like um you know when when the
nest mates come to the surface of the
ground at in the morning just as the sun
is
Dawning we don't know how they're
sending El you know information within
the colony but it's like they're
predicting that it's about to be the
right time to
forage so it's or or whatever this thing
is we don't know what it is but it's
acting like it's predicting what's
happening
great thanks the ordering is it's it
there's a lot
of it's like the the um slide slide some

game I forget what it's called but it's
like where somebody else makes the
slides for you and then you have to do
like a lightning
talk slides in a random order yeah yeah
because I I definitely when I looked at
it the first time I definitely read it
as a subway map but yeah I when I looked
at it the more I was like a bit confused
about the motivation between putting it
like
that here were some of the awesome
slides from Maria with the history of
perception and about how how it's it's
an
ancient from from
Antiquity to to the the uh the idea that
pre-existing information structures
sensory
data then where the book kind of picks
up is with
helmholtz and said and that's sort of
this like hypothesis
testing then that sort of like
hypothesis testing
idea finds a more natural fit with
basian
epistemology rather than for example
example scientific positivism or
scientific
falsification approach so it's not that
we're just like seeking to lock in a
fact about observation or seek to lock
in the rejection of a hypothesis it's
more that there's just this like
envelope or portfolio or distribution
that's getting updated and so that
aligns more closely with the basian

methods than with um kind of rule it in
in positivism or rule it out
falsificationism
okay 3.2 I think it'd be a cool thing if
we uh maybe now it'd be easier we could
paste it into an llm and just say
generate the code for this but it's just
showing that you can have um chains of
partial
correlations like of everybody where a
happens 50% B happens of everybody where
B happens 50% C happens what percentage
of a had C happen and then you can just
multiply the probabilities like a
chain if one defines preferred States as
expected States this the book then one
can say living organisms must minimize
the surprise of their sensory
observations I'm not clear on the
ontology here is it that preferen is the
same as expectation or formally we can
use them in the same equation or
something else does active inference
make ontological claims EG preference is
actually expectation just as heat is
actually exitation of
molecules more generally is free energy
in physics just analogy or an
ontological
assertion this was brought really nicely
together in in
um I'm forgetting the first um author
but Stephen Francis
man
p as a measure of both probabilities and
preferences that's where we get
pragmatic value that's the clearest
difference from reinforcement learning

rather than the proposal of a totally
ancillary secondary reward
distribution that's like observations or
observations plus actions like Q
learning rather than projecting those
onto a
reward ranking and then using that
reward ranking in this actionable
way the observations directly are going
to be compared with a preference
distribution which is a prior over
sensory observations and that does the
pragmatic value
work but some and then they they kind of
even talk about how like people will
lean on one of those interpretations or
another but they're they're both equally
viable
is there a minimal example of a thing
defined by marov
blanket such as a cell or or some
subcellular
elements is a thing Persistence of marov
blanket are all things non- equilibrium
steady state
processes do we need to be more strict
in saying these things are only self-
evidencing things rather than things in
the colloquial
sense some work towards that is is from
um particular kinds path integrals
particular kind strange things
paper just showing that um
ranging from inert
things to classical simple
things to strange and self-reflective
things they all fit within this
interface concept it's really just what

is the modeler system of interest and
that's also the first step of the
chapter 6 recipe for modeling and it's
non-trivial to identify the system of
Interest I mean if you're studying the
heat transfer from the
engine that you could make an argument
that the system of inter should be like
the solar system or you could say okay
but I'm only interested in the
99% heat um transfer threshold and then
at that point you might even exclude
some of the plastic on the engine so
it's not that there's like one
um kind of thing or or maybe you're
interested in a specific thingness
I'm reminded of um fields's course as
well Chris fields's
course like in in
how uh the physics of of information
processing right and how you have to
define a thing and and really going down
to um bits being the fundamental
construct of the
Universe um you know in
um um
shoot I've got I've got the I've got the
article on my bookshelf behind me but um
art wheeler you know John Wheeler John
archal Wheeler it from bit right so you
know ultimately things are constructed
from um plank units right
so yeah and the bit is like a collapse
onto a yes or no question that's like
the simplest
question okay I'm even going to look at
what these are
hylomorphism every physical object is a

compound of matter and
form hylozoism Doctrine all matter has
life okay
what is the analogy to friction in a
cognitive system how strong is the
analogy okay nothing here what would
anyone
say like if we're thinking okay the
baseball is going to go in the parabola
path of least
action baseball's quite massive compared
to the air molecules still though there
is friction with the air some of that
energy is going to be lost to
heat and then if it were in a vacuum you
would lose less energy to heat if it
were in a very thick
liquid you would lose more energy to
heat so this person is asking about
how these these equations might work
well for like the frictionless
Assumption like um you know it's a
collision on the frictionless billiard
table and no energy is lost from hitting
the sidewalls and then you get a certain
play out but then if you do have
friction on the table it's like
everything is getting slowed down
because it's getting dissipated what do
you think Jeff I think the distribution
curve is going to be wider right I mean
if if an agent is
hungry
or um
dehydrated or any other type of um state
that
impairs um a optimal band of processing
capacity then you're going to have a

broader yeah you're going to have a
broader distribution
curve like if I'm under the influence of
drugs you know um that
you
know make me
hallucinate my ability to interact with
my environment is going to be diminished
right
but but isn't isn't friction just
included in the oil garage questions
like isn't that just shaping the the
landscape of which you find the the path
of leas resistance
like it's about finding the equations of
motions right and if friction is part of
your system that
is going to be part of of finding
finding the
PA maybe with very high friction does it
break down
also friction is so close to friston
it's a classic
typo it's so it's so funny also just
reading these like okay then there's the
helm
holes but we just saw Helm Holts from
the perception as unconscious
inference side
but then here's Helm holtz's work from
the um
Flow dynamical
side so it's like it was
already right
there you can modify the oiler lrange
equation by adding nonzero right hand
side to incorporate friction that's
dissipation function

okay let's let's keep it in mind one one
thought about the under the influence
are there any treatments or or or
substances that reduce the variance on
performance or
are is is or do do
interventions always are usually
increase
variance like if you you did something
just quote normally a 100
times and then okay now do it with one
eye closed or do it like you know
standing on one leg or do it when you're
really
tired W maybe some of those or or do it
on caffeine but it's like it might be
you might be better you might be worse
on average but then would any of those
have a tighter
distribution or when we change something
do we always make it more variable too
seems like it will be very oh sorry go
ahead Jeff no go ahead go
ahead seems like it will be very like
dependent on the on the task um as well
but I'm sure that
certain interventions would lead to both
higher and and and lower variance like I
don't know if you would take
a sleeping pill that would like reduce
the the bar of what you'll be doing the
the next the next few hours but uh we
probably increase the variance in your
performance on on certain tasks like I
don't know reading a book or doing
something that requires you to be
awake I'm coming down to instrumentation
and

augmentation
um if I'm you know in my natural state
in my body I'm not going to be able to
stay underwater and explore for a very
long time but if I have a
a mask and a snorkel then I'm going to
be more adapted to the environment but
it is a technological adaptation not a
biological ad adaptation and if I have a
SCUBA tank even more right so that is
tightening up the curve in terms of my
performance in that
environment yeah that's a great Point
too is the the off Source I mean if you
go on a ventilat
or a pacemaker you could have lower
variability in your heart rate or
breathing or if you had an exoskeleton
that was like drawing for you you you
would have lower
variability but with kind of
cascading
externality and this comes back to me
again to the to the fields course where
you know we've got this mesoscale where
we're naturally inclined to be able to
understand and comprehend but when you
look at you know things at the you know
below the the classical Newtonian scale
if you look at astrophysics or quantum
physics um there is
no conjoining or comprehension without
instrumentation so so we literally can't
even Explore those environments without
augmentation yeah well it's like you you
all like uh let's just say the base
balls are flying around in the dark room
we can just put on a strobe light take a

picture we can say well the photons of
our strobe light probably didn't
influence the baseball that much that's
why it's classical but then yes when you
get into things that are on that scale
all of the sudden it's like it goes from
not mattering not mattering barely
barely barely barely barely and then all
of a sudden it's like it
overwhelms when you're studying things
are at that exact scale then your
instrument becomes as big it's like your
thermometer has a certain heat capacity
if you dip it into a little to one
molecule of water you'd have in order to
not in order to get an accurate read on
that temperature you would have had to
already know the actual
temperature otherwise you would be
getting a compromise between the
temperature that it really was and what
you thought it was but you would yeah it
it's funny because I was just reading
about from economics I was reading about
jeevan's paradox
which is that the more efficient a
becomes the more or the more the more
efficient um you become in resource
extraction from an environment the more
you're going to use that resarch that
that that resource and then it could
cause it to collapse so like if I'm you
know I'm I think an example is um you
know the EPA requires higher fuel
efficiency on cars so because of the
higher fuel effic efficiency um you uh
people wind up driving more because it's
cheaper to drive and you're actually

putting out more pollutants into the
environment than you would have
otherwise and then another car related
one would be like adding a lane to the
freeway and then people think oh great
now there's more lanes and then oh yeah
capacity planning for Highway it's huge
right like like you look at Houston or
or anywhere where they do widening and
it just stays it just stays it's
actually that they're incentivizing
people to get in cars more um and and
not realizing that it's going to cause
you know friction or eventually collapse
right you know in the end it you know
it's it's it's collapse Building Systems
that have uh like like if you look at
how the the the camir civilization fell
it was because they created all this
wonderful um
um water um Management systems and and
it was very very brittle and as soon as
they had a drought or a flood then it
like it they didn't have water you know
and then people literally just died from
dehydration so and the other thing from
a car perspective right we can look at
it at many scales me driving my car
independently doesn't do much in terms
of pollution right it's not really
measurable in terms of the scale of of
of the earth B biosphere but everybody
driving their cars has absolutely a
measurable
impact thank you all so much see you
next time I've got another Peace Jeff
Frank you wna bring up anything
H last point on kind of the friction I

mean considering friction as energy that
is used that that becomes converted to
entropy and not like to work if we
consider statistical work to be
Improvement in accuracy
and then the entropy is just the
variance estimator which could be
overfit or it could be
inflated so then it's
like if a new observation comes in the
sort of um 100% efficiency motor it
would take in one calorie of energy and
it would translate it to one calorie of
moving the car forward so that' be 100%
efficiency it wouldn't change the
temperature of the environment at all
whereas if you just burnt the calorie
the car would go nowhere and you would
have released one calorie perfectly as
heat so there's probably some
connections there with with the
information content of of what the model
ingesting and then it's like is that
information getting like cleanly burnt
into the best
possible ACC accuracy
Improvement or is it just being like
thrown
away but then it's like what would it
mean to throw it away like if it just
what didn't pay attention to the new
data point and it just deleted the file
in the real
world that would change the heat balance
in the real world because of like the
energy and and um information
equivalences but I don't know if that's
exactly the right angle

there okay
well August is gone
almost if you return at this time next
week it'll be chapter
eight uh oh let's see this time next
week yeah chapter 8 part two which um
everyone's welcome to
join otherwise wise there'll be another
three and then we uh are on with it but
any
last comments or
questions oh just say yeah just thank
you and see you next time cool thank you
Frankie yeah thank you yeah thanks a lot
Daniel bye talk to